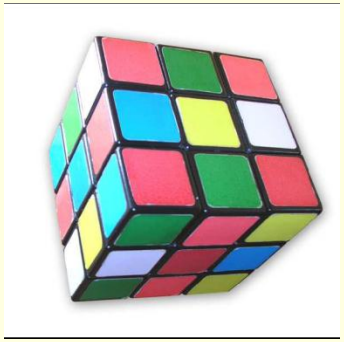
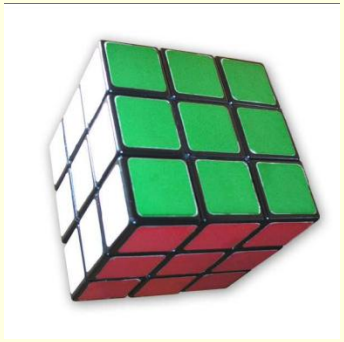




Artificial Intelligence
(Part 2)
Knowledge Representation and Search

Course Contents

Again..Selected topics for our course. Covering all of AI is impossible!

Key topics include:

- Introduction to Artificial Intelligence (AI)
- Knowledge Representation and Search
- Introduction to AI Programming
- Problem Solving Using Search
- Exhaustive Search Algorithm
- Heuristic Search
- Techniques and Mechanisms of Search Algorithm
- Knowledge Representation Issues and Concepts
- Strong Method Problem Solving
- Reasoning in Uncertain Situations
- Soft Computing and Machine Learning

Basic AI Building Blocks

- Ability to represent *knowledge*
- Problem solving by *searching*
- Ability to *reason*
- ...Many more...
- AI programming languages

AI as representation and search

Artificial Intelligence :- study of representation and search through which activity can be performed on a mechanical device (engineering perspective)

Knowledge Representation

Example: given the headline "*Najib wins election".* Could a machine answer the question "*Who is prime minister?"*

- Background knowledge is necessary
- How is this knowledge written down, or encoded so a computer can use it?
- How can it be written down efficiently? We can't write everything down.
- What do the formal representations mean? Semantics.

But What is Knowledge?

- Rules?
  - Production systems
- Classification?
  - Semantic networks
- Structure?
  - Frames
- Idealised and formalised knowledge?
  - Logic

Why need representation?

-capture important features of a problem

-make information accessible to problem solving procedure

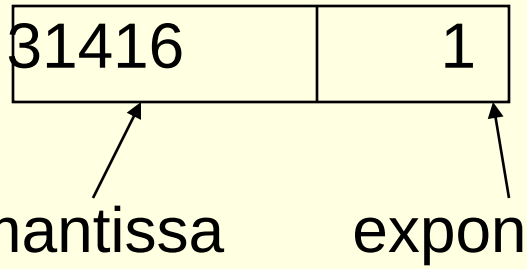
-abstraction

e.g.

The real number: π

Decimal : 3.1415927….

Floating point :



Computer memory: 11100010

AI as representation

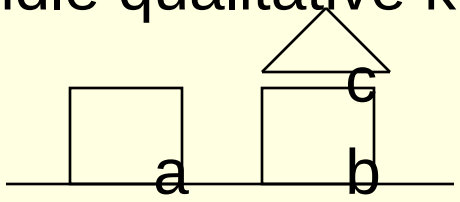
A representational scheme should:

1. Be adequate to express all of the necessary information
2. Support efficient execution of the resulting code
3. Provide a natural scheme for expressing the required knowledge

AI as representation

To support all these needs, AI representation language must:

1. Handle qualitative knowledge-(reasoning)



clear(c),clear(a), ontable(a), ontable(b)

on(c,b), cube(a), cube(b), pyramid(c)

Predicates : clear, ontable, on, cube, pyramid

AI as representation

2. Allow new knowledge to be inferred from set of facts and rules

$\forall X \neg \exists Y$ on(X,Y) ➔ clear (X)

-for all X, X is clear if there does not exist a Y such that Y is on X-

3. Allow representation of general principles as well as specific solutions

4. Capture complex semantic meaning

hassize(bluebird, small), hascovering(bird,feathers), hascolor(bluebird,blue), hasproperty(bird,flies)

5. Allow for meta-level reasoning

"knowing about what you know" – meta-knowledge

"Meta-" is used to designate something that applies to the thing as a whole, often including itself. For instance, a meta-language is a language that used to talk about language.

AI as representation

Knowledge representation languages:

Predicate calculus, semantic networks,
frames, objects, rules,..etc.


Knowledge and Intelligence
• Does knowledge have any role in demonstrating intelligent behavior?
Sensing
Decision maker
Knowledge
Action

AI as representation and search

Problem solving as search:

Problem are solved by searching among alternative choices, is supported by a commonsense view of human problem solving.

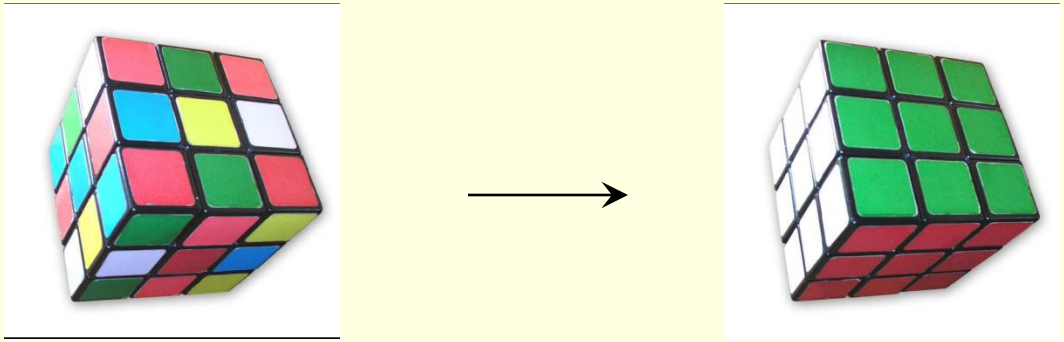
Search

- Often no direct way to solve a problem.
- You may know what moves are allowed but not *how* to put the moves into a sequence to solve a problem.
- Can generate possibilities for next step and so on.
- Considering full search space often too expensive. Too many possibilities (even for computers).

Search Examples

- Planning a journey
  - Route finders
  - Timetabling
- Games
  - Noughts and crosses
  - Chess
- Puzzles

Rubik's Cube



- 43,252,003,274,489,856,000 combinations
- Up to 481,229,803,398,374,426,442,198,455,156,73 brute-force solution attempts
  - More than 15,259,696,962,150,381 years!!
- Need to look at heuristics or strategies, i.e. selecting the best options to lead to a solution.

Reasoning and Inference

- If we know that elephants are mammals with four legs and that Barbar is an elephant, can we conclude that Barbar is a mammal with four legs?
- How do we deal with an elephant (and a mammal) but only has three legs?
- If we use formal *logic* as a knowledge representation language, logical proof can be used to allow us to infer new facts.

AI Programming Languages

- Need a language good for symbolic manipulation (rather than numeric).
- LISP - short for List Processing. A functional programming language. Many AI systems have been written in LISP.
- Prolog - we'll use this language for the course. Short for programming in logic. Based on a proof method in logic called resolution. Easy to get started and good for symbolic manipulation.
- But can be written in any general purpose language, e.g. C or Java. Particularly, as AI programs become components of other larger systems rather than stand alone systems.

Intelligent activity is achieved through the use of:

1. Symbol or pattern to represent significant aspects of a problem domain
2. Operations on these patterns to generate potential solutions to problems
3. Search to select a solution from among these possibilities

Next…

- Propositional calculus
- Predicate calculus